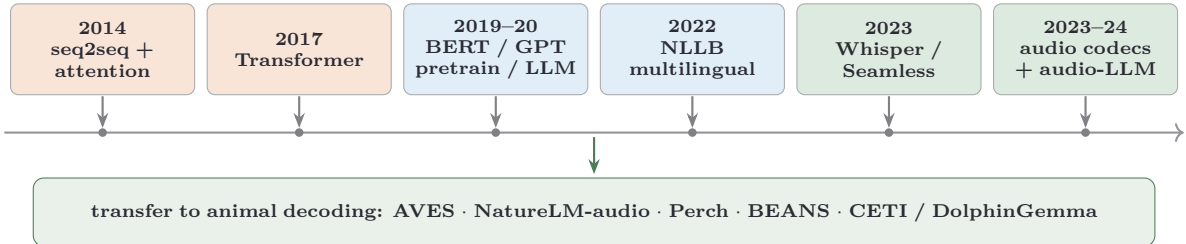
2014
seq2seq +
attention
2017
Transformer
2019–20
BERT / GPT
pretrain / LLM
2022
NLLB
multilingual
2023
Whisper /
Seamless
2023–24
audio codecs
+ audio-LLM
transfer to animal decoding: AVES · NatureLM-audio · Perch · BEANS · CETI / DolphinGemma